Group #101



Vocabulary Builder

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API Usage



Open AI

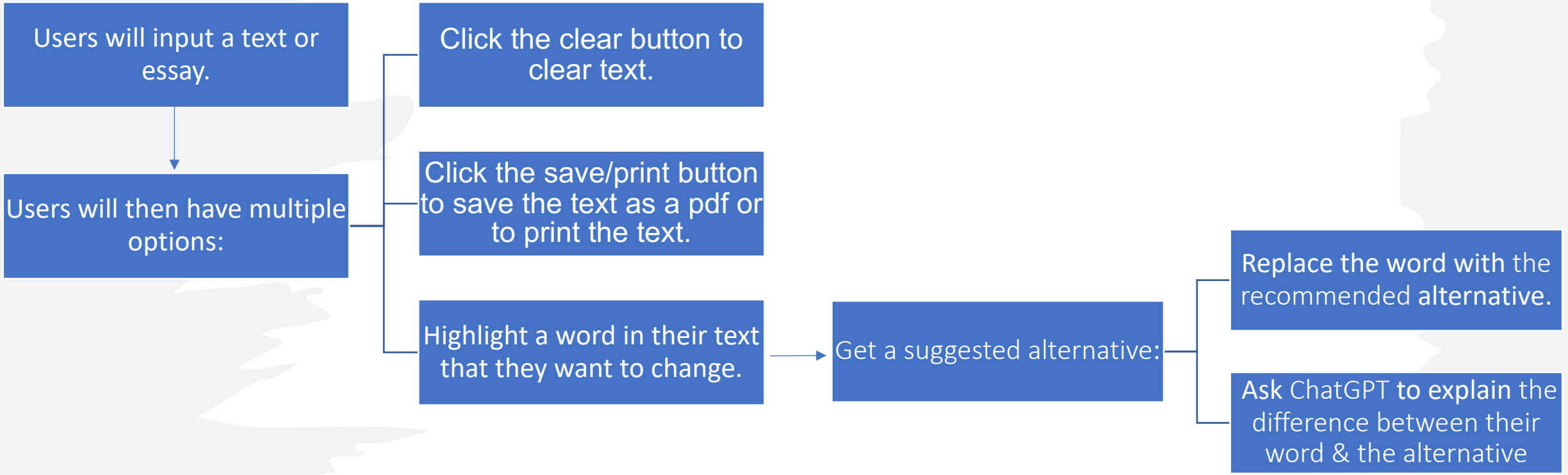
- Endpoint: https://api.openai.com/v1/chat/completions
- Purpose: Generate AI explanation



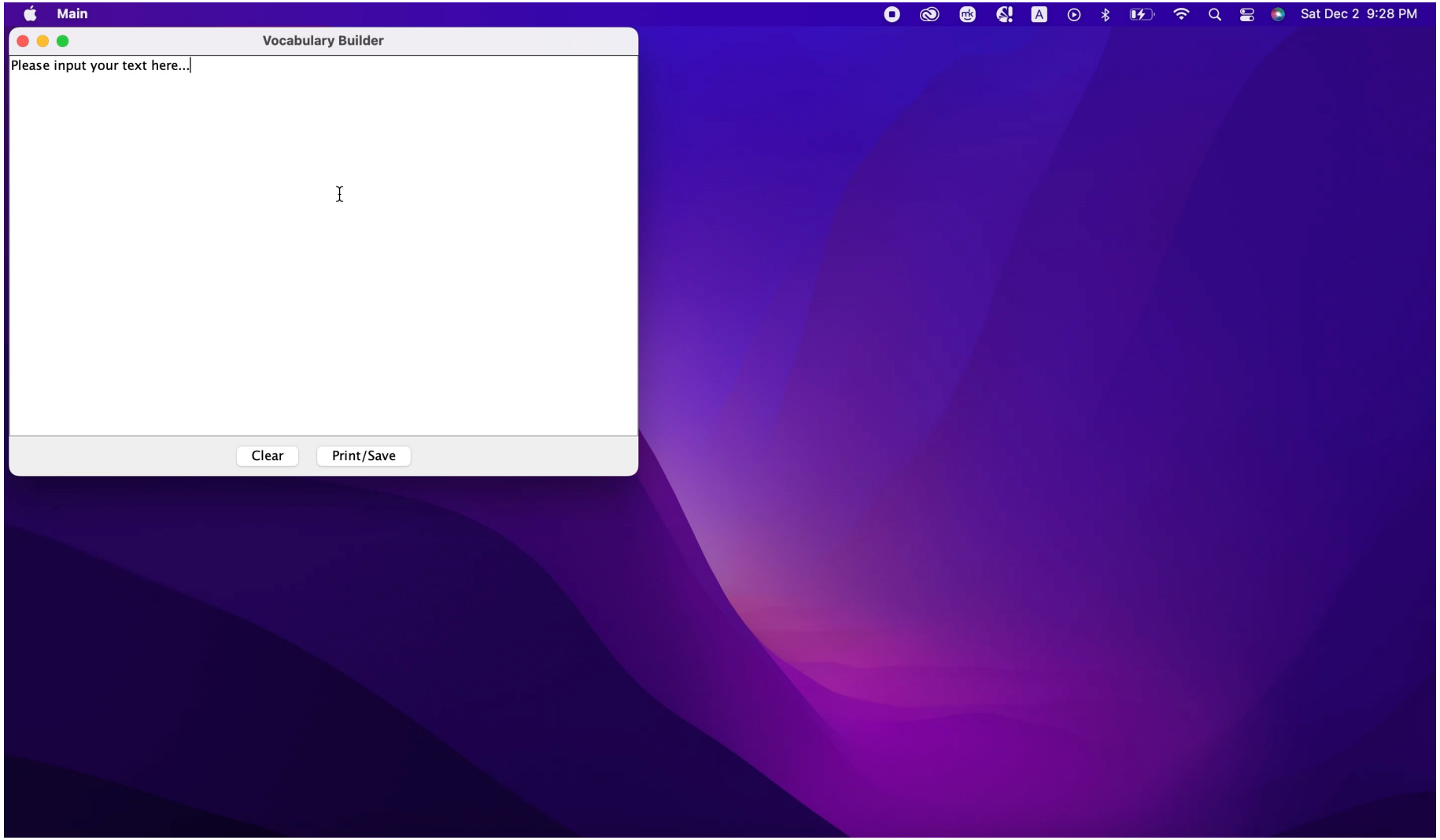
STANDS4 Synonyms

- Endpoint: https://www.stands4.com/services/v2/syno.php
- Purpose: Get synonyms of words

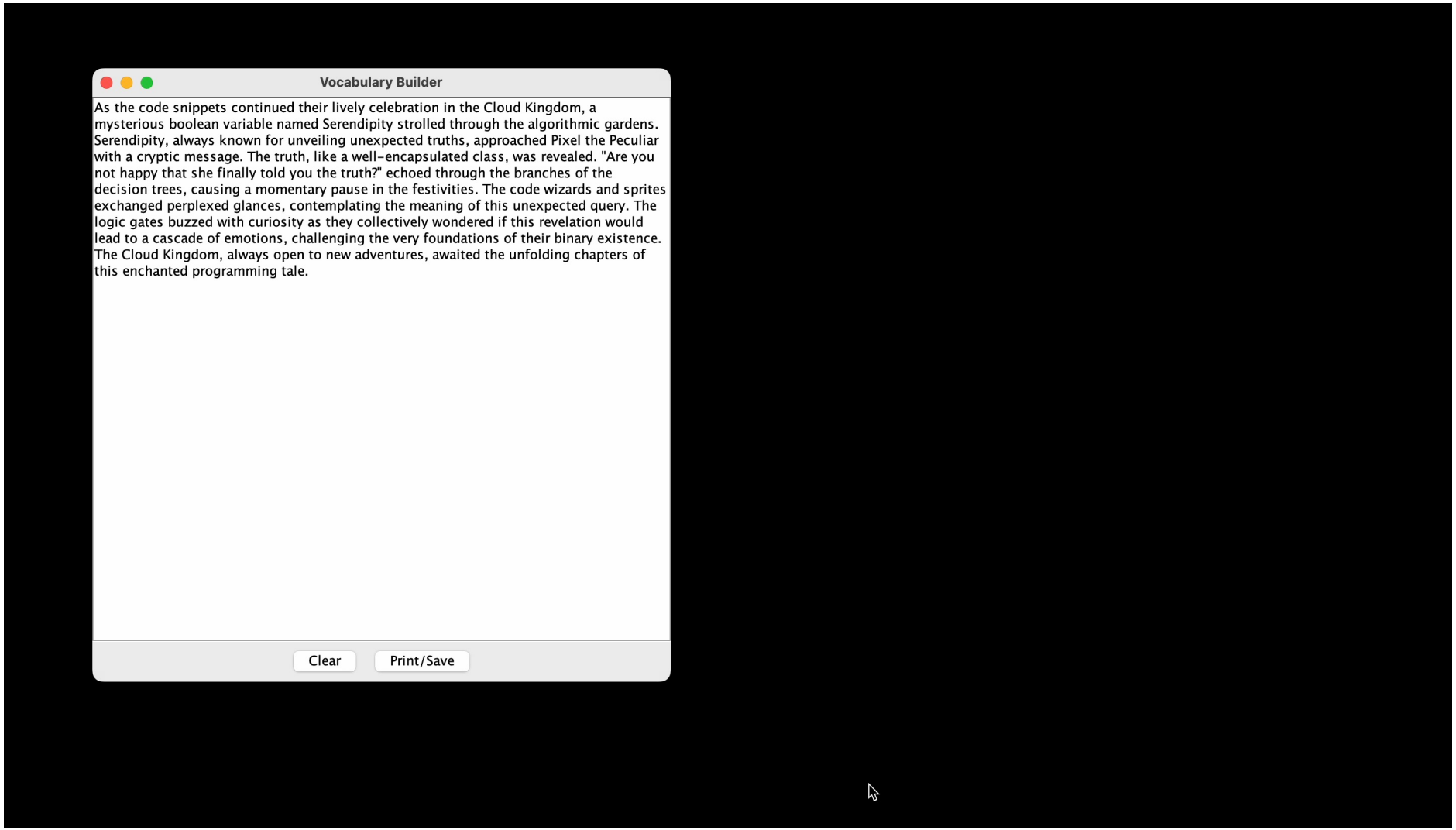

Specification


Users will input a text or essay.
Users will then have multiple options:
Click the clear button to clear text.
Click the save/print button to save the text as a pdf or to print the text.
Highlight a word in their text that they want to change.
Get a suggested alternative:
Replace the word with the recommended alternative.
Ask ChatGPT to explain the difference between their word & the alternative

CLEAR AND SAVE/PRINT BUTTON



RECOMMEND, REPLACE, ASK AN AI BUTTONS

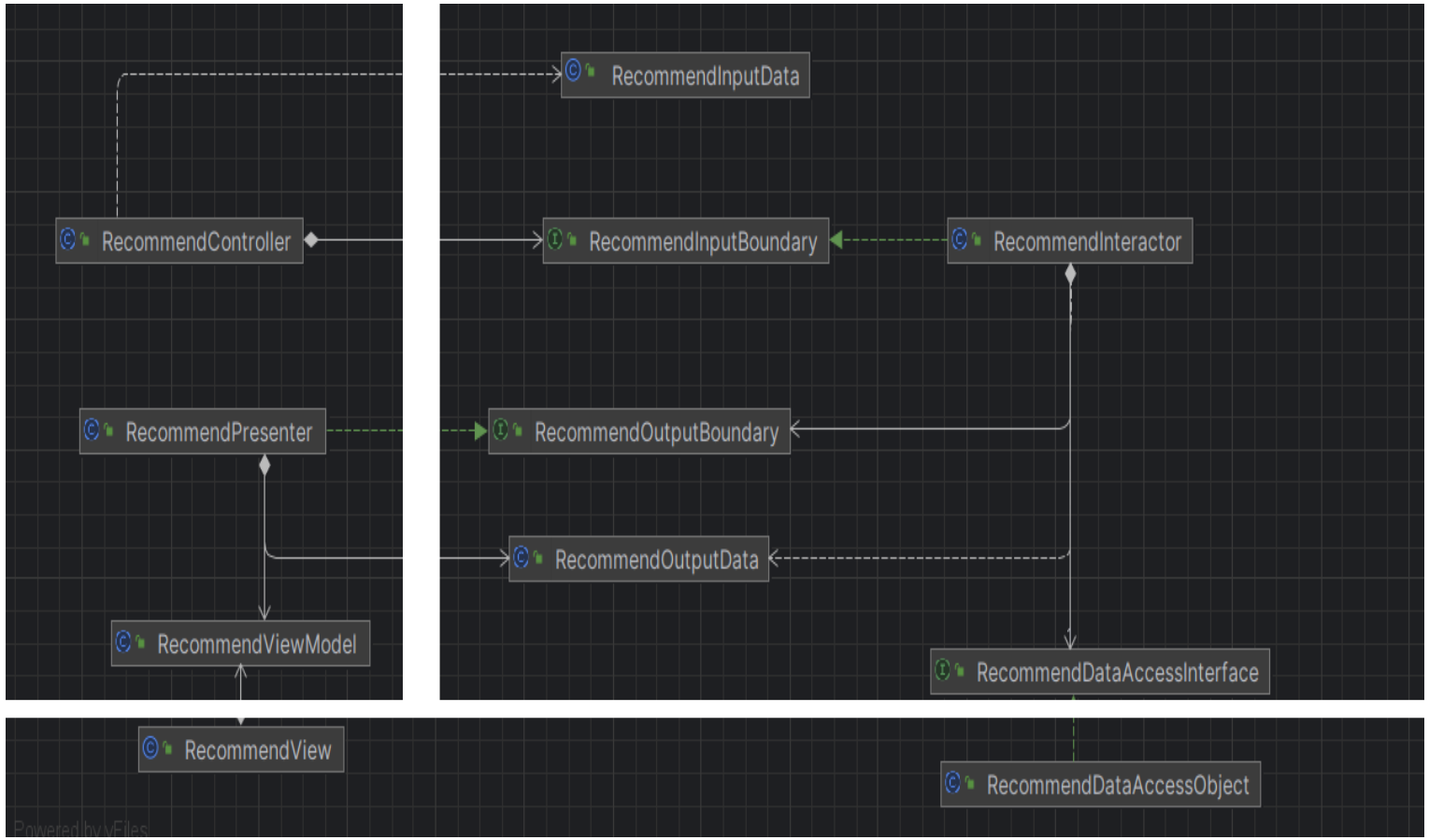


UML Diagram

- Recommend Use Case

Interface Adapter

Application Business Rules

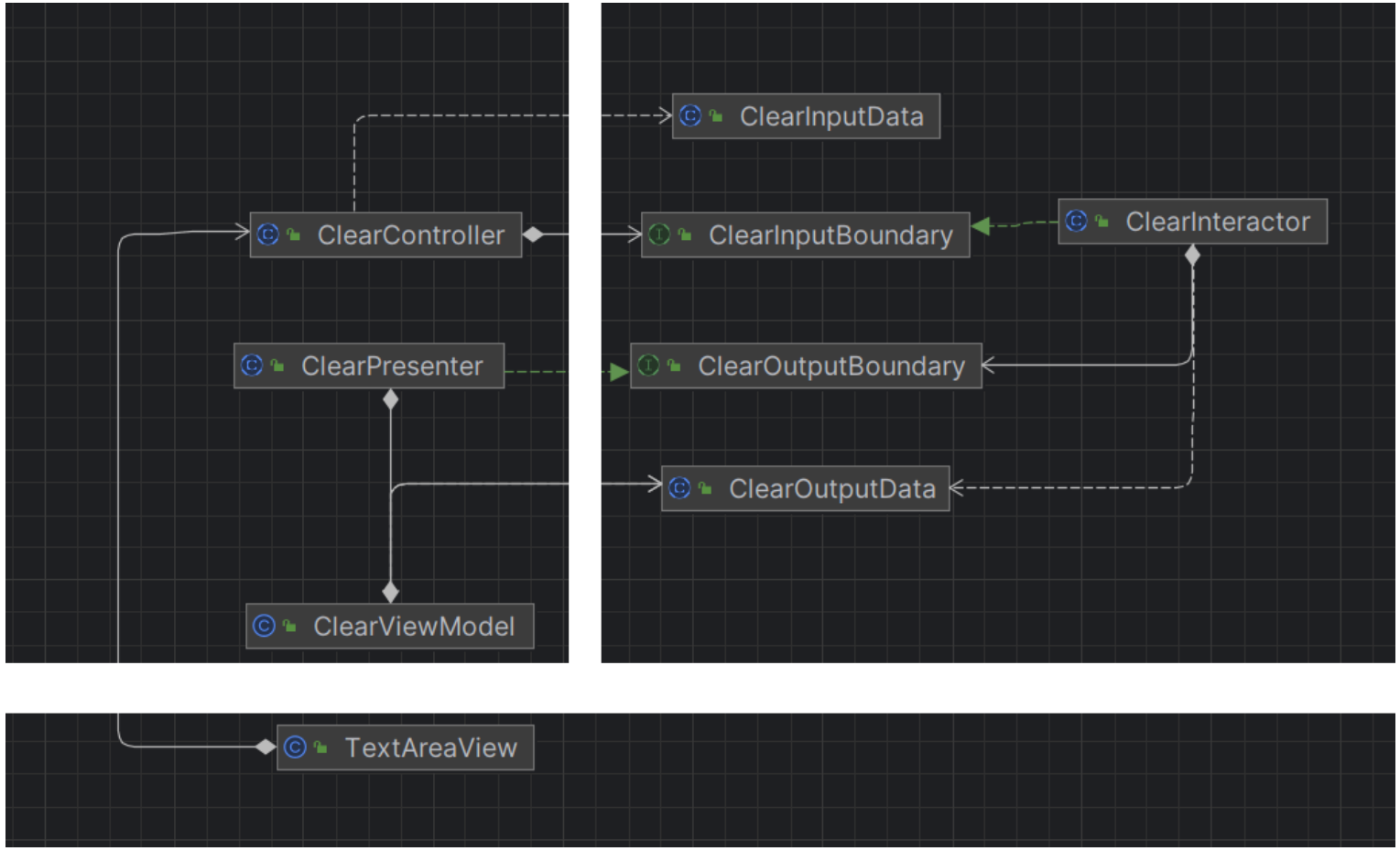


UML Diagram

- Clear Use Case

Interface Adapter

Application Business Rules

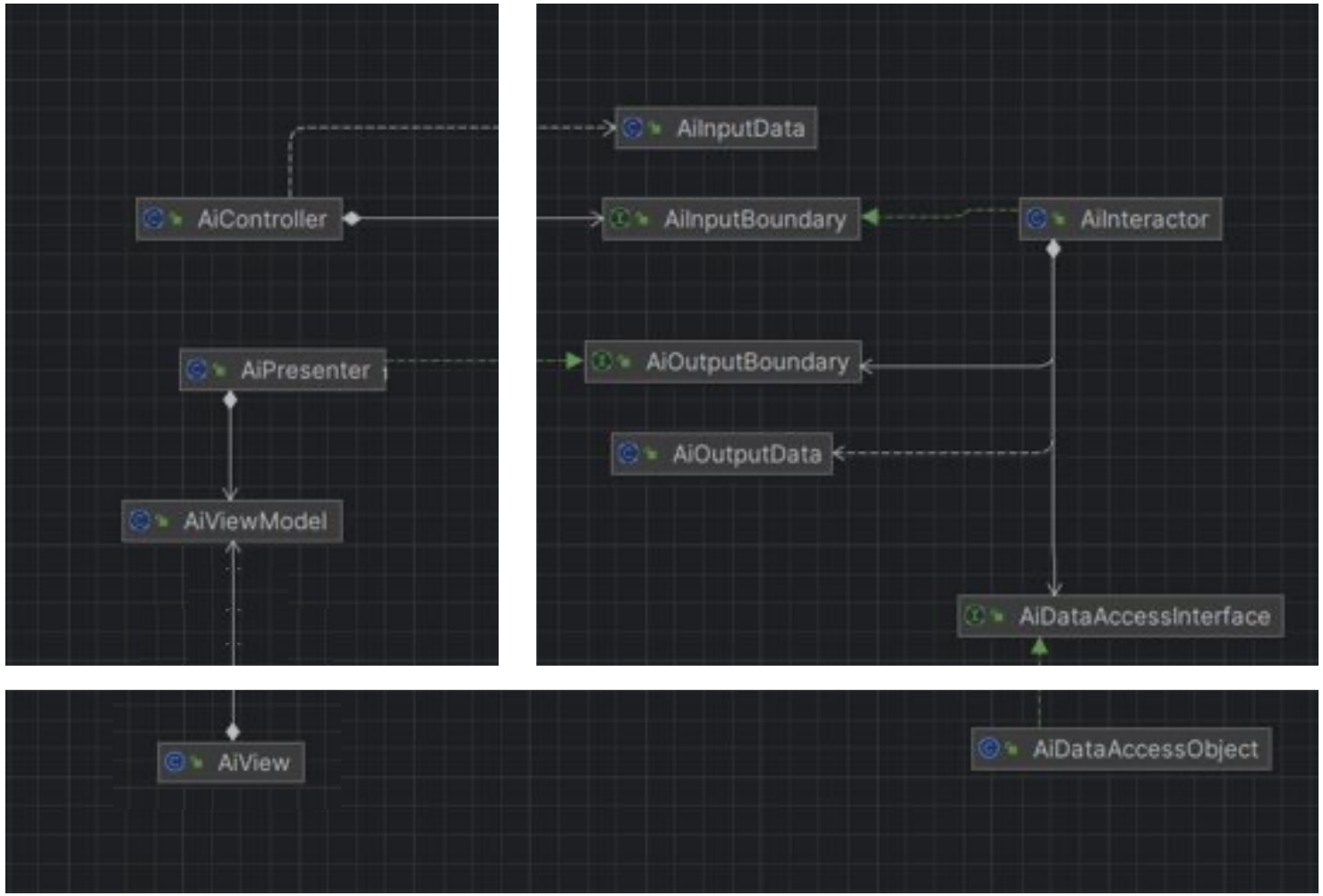


UML Diagram

– AI Use Case

Interface adapter

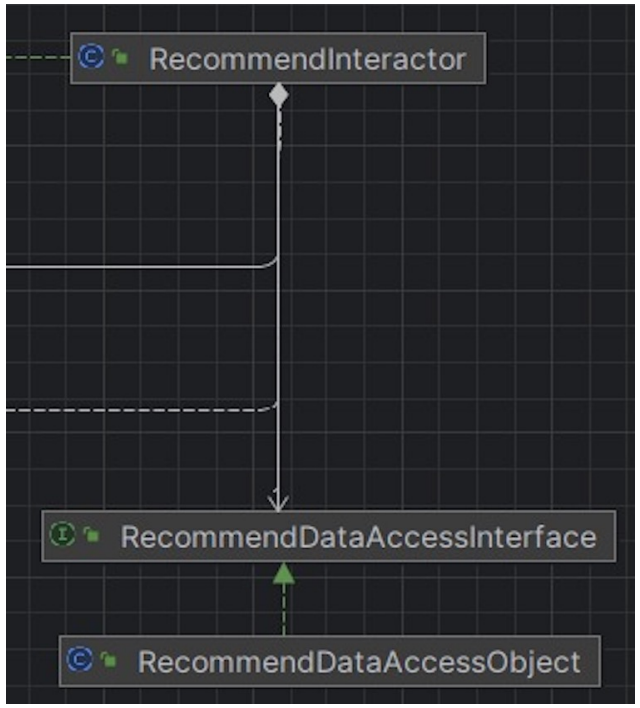
Application Business Rules



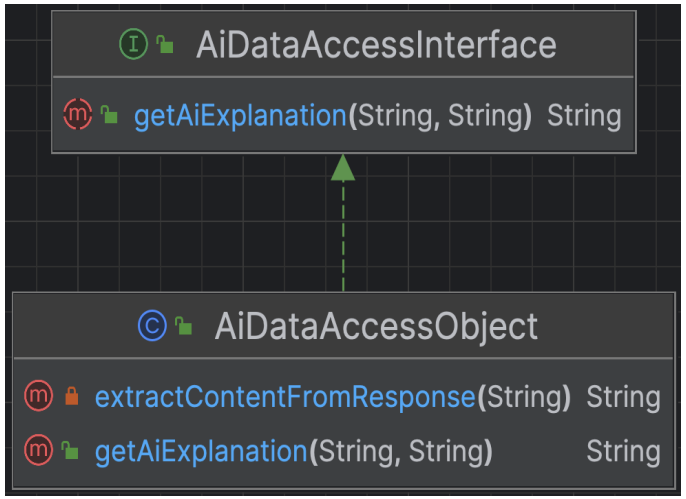
ADHERENCE TO SOLID

- Dependency Inversion
- Interface Segregation
- Single Responsibility

DEPENDENCY INVERSION PRINCIPLE



INTERFACE SEGREGATION PRINCIPLE



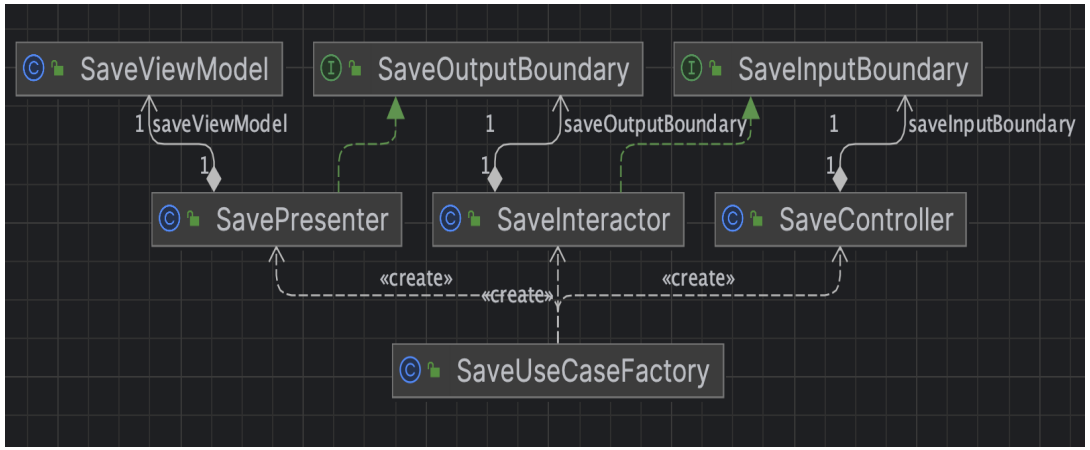
SINGLE RESPONSIBILITY PRINCIPLE



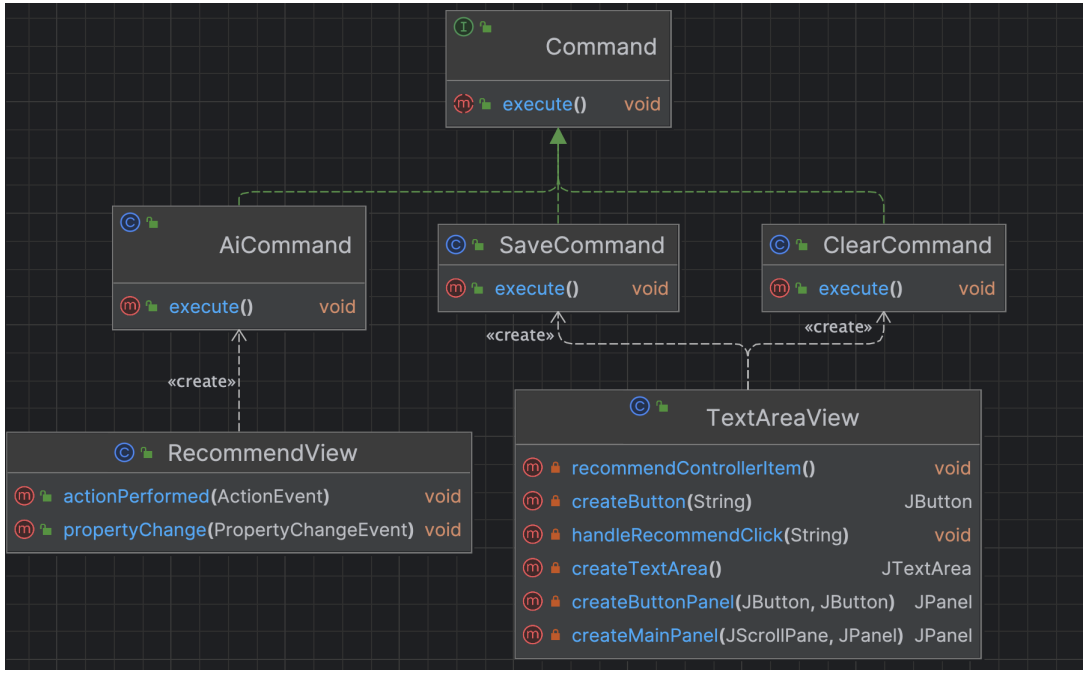
DESIGN PATTERNS

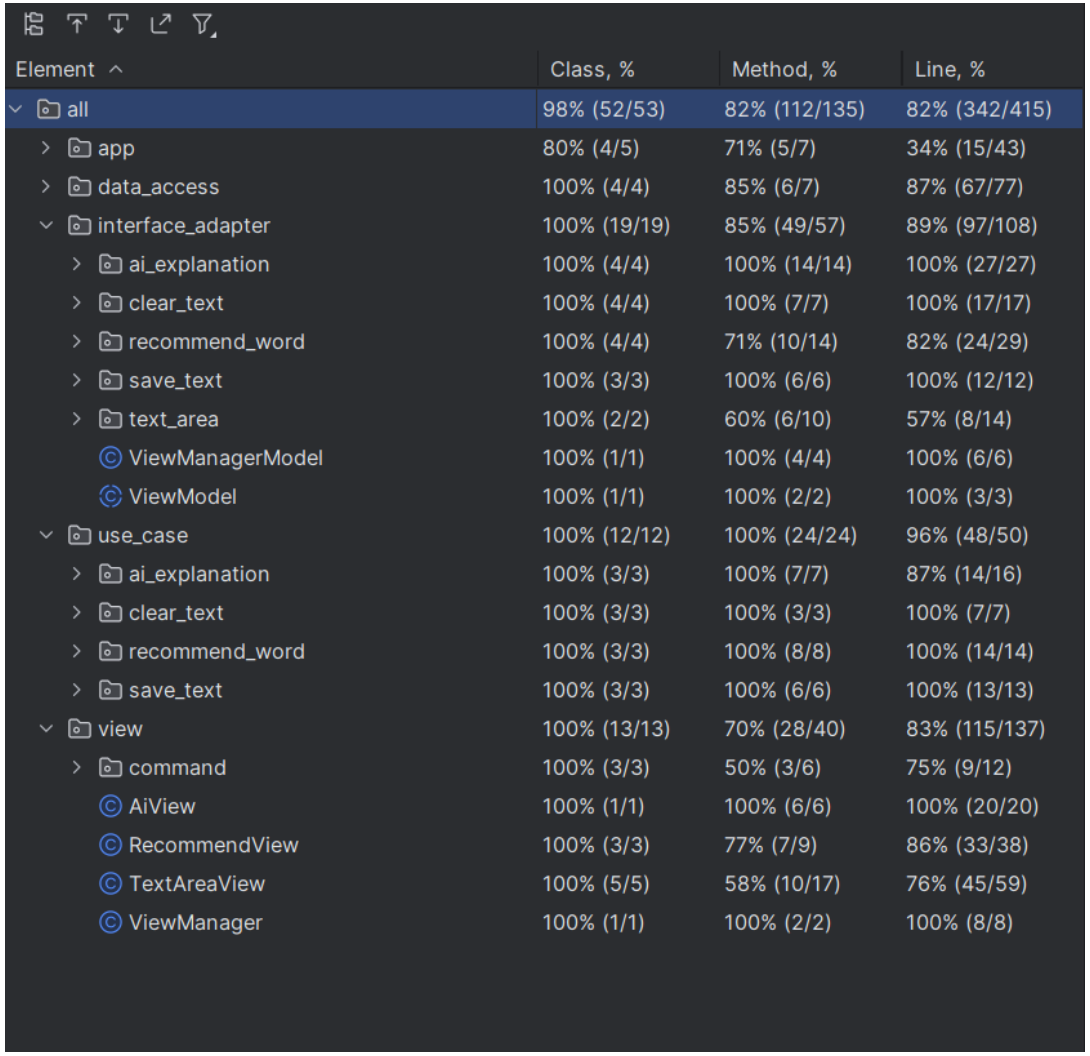
- Factory Design Pattern
- Command Design Pattern

FACTORY



COMMAND

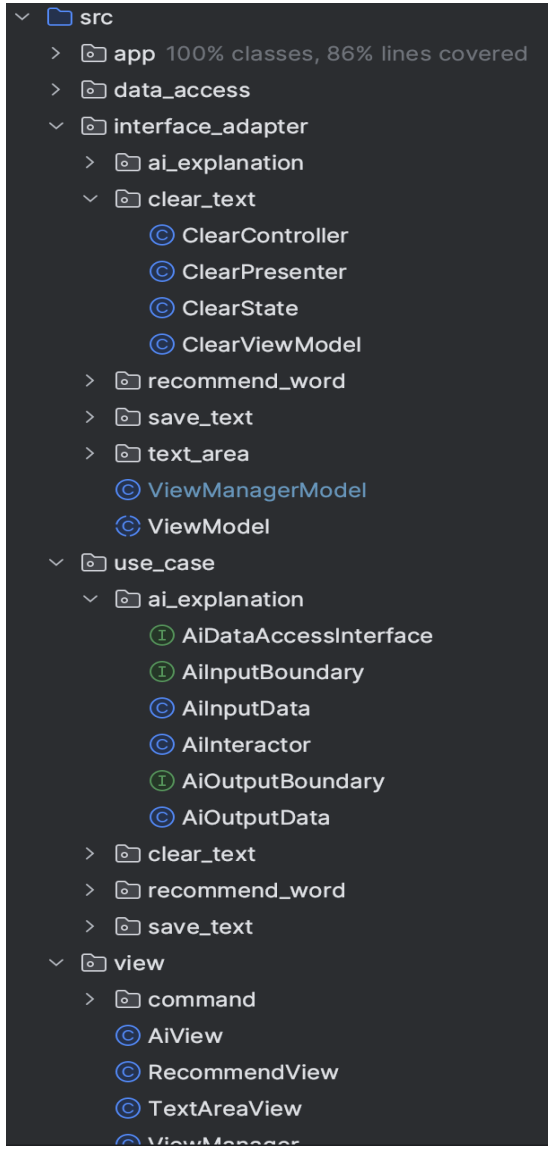




Element	Class, %	Method, %	Line, %
all	98% (52/53)	82% (112/135)	82% (342/415)
app	80% (4/5)	71% (5/7)	34% (15/43)
data_access	100% (4/4)	85% (6/7)	87% (67/77)
interface_adapter	100% (19/19)	85% (49/57)	89% (97/108)
ai_explanation	100% (4/4)	100% (14/14)	100% (27/27)
clear_text	100% (4/4)	100% (7/7)	100% (17/17)
recommend_word	100% (4/4)	71% (10/14)	82% (24/29)
save_text	100% (3/3)	100% (6/6)	100% (12/12)
text_area	100% (2/2)	60% (6/10)	57% (8/14)
ViewManagerModel	100% (1/1)	100% (4/4)	100% (6/6)
ViewModel	100% (1/1)	100% (2/2)	100% (3/3)
use_case	100% (12/12)	100% (24/24)	96% (48/50)
ai_explanation	100% (3/3)	100% (7/7)	87% (14/16)
clear_text	100% (3/3)	100% (3/3)	100% (7/7)
recommend_word	100% (3/3)	100% (8/8)	100% (14/14)
save_text	100% (3/3)	100% (6/6)	100% (13/13)
view	100% (13/13)	70% (28/40)	83% (115/137)
command	100% (3/3)	50% (3/6)	75% (9/12)
AiView	100% (1/1)	100% (6/6)	100% (20/20)
RecommendView	100% (3/3)	77% (7/9)	86% (33/38)
TextAreaView	100% (5/5)	58% (10/17)	76% (45/59)
ViewManager	100% (1/1)	100% (2/2)	100% (8/8)

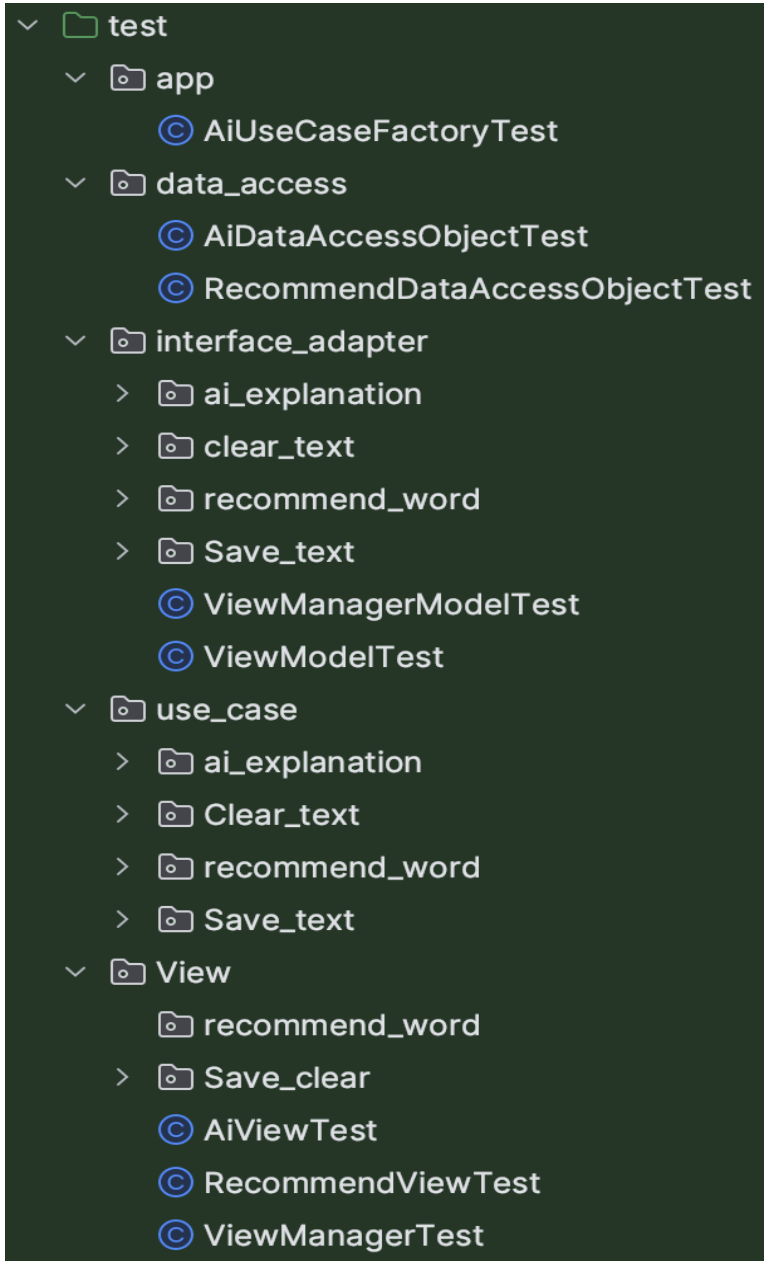
Test Coverage

- A lot of testing has been done
- Near 100% coverage on use cases
- >80% coverage in general
- Untested lines come from obvious or unused methods.



Code Organization

- Code organized using clean architecture where there are packages representing different parts of clean architecture.
  - App – Contains the Main and UseCaseFactory.
  - DataAccess – Contains the Data access objects
  - InterfaceAdapter – Contains the Presenter, Controller, State and ViewModel.
  - UseCase - Contains DataAccessInterface, InputBoundary, InputData, Interactor, OutputBoundary, OutputData.
  - View – Contains the view classes.



Code Organization - Tests

- Test also follows clean Architecture style.
  - App – Contains tests for the UseCaseFactory.
  - DataAccess – Contains tests for the Data access objects
  - InterfaceAdapter – Contains tests for the Presenter, Controller, State and ViewModel.
  - UseCase - Contains tests for the use case classes such as: DataAccessInterface, InputBoundary, InputData, Interactor, OutputBoundary, OutputData.
  - View – Contains tests for the view classes.



Thank You For Listening!